

Andrew Denry Price

Senior Data Engineer

Tampa, FL 33618

andrewprice0713@outlook.com

+1 813-563-5553

<https://www.linkedin.com/in/andrew-d-price-09a9793b1/>

Summary

Senior Data Engineer with 11+ years of experience designing and delivering HIPAA-compliant, cloud-native data platforms for healthcare and enterprise environments. Proven expertise in building scalable batch and streaming pipelines, modern lakehouse architectures, and analytics platforms that reduce latency, optimize costs, and enable clinical and operational decision-making. Strong leader with a track record of mentoring engineers and partnering with stakeholders across engineering, analytics, and compliance.

SKILLS & COMPETENCIES

- Python (NumPy, Pandas)
- Java, Scala
- R
- C++, C#
- SQL (advanced query optimization)
- **AWS:** Lambda, S3, API Gateway, DynamoDB, EC2, Glue, Redshift, EMR
- **Azure**
- **GCP:** BigQuery, Dataflow
- Oracle Cloud, IBM Cloud
- ETL / ELT Pipelines
- SAP
- Apache Spark, Spark ML
- Kafka, Flink, Storm
- Hadoop, Hive, MapReduce
- Airflow (MWAA)
- Feature Pipelines & Real-time Streaming
- Databricks
- dbt
- Microsoft Fabric
- Snowflake
- BigQuery
- MySQL, PostgreSQL
- MongoDB, Cassandra
- Redis, Elasticsearch
- Power BI
- Tableau
- Looker, LookML
- Qlik
- Kibana
- HIPAA Compliance
- FHIR
- Epic EHR
- Data Governance & Security Controls
- MLflow
- SageMaker
- Model Monitoring
- MLOps Data Workflows
- Docker, Kubernetes
- CI/CD Pipelines
- Agile / Scrum

Education

University of Florida

Bachelor's degree Computer Science 3.76

2010 – 2014

WORK EXPERIENCE

Senior Data Engineer

Aug 2021 - Present

Optum

Responsibilities:

- Designed and maintained cloud-native healthcare data pipelines supporting analytics, reporting, and predictive modeling across multiple clinical domains.
- Led the architecture and delivery of a lakehouse platform, reducing infrastructure and storage costs by 60% while improving query performance and scalability.
- Built real-time and near-real-time ingestion pipelines using Apache Kafka and Flink, reducing end-to-end data latency by 25%.
- Improved overall data platform efficiency by 40% through pipeline refactoring, Spark tuning, and optimized partitioning strategies.
- Developed scalable ingestion frameworks handling high-volume, multi-source healthcare data, increasing processing throughput by 29%.
- Implemented new ETL/ELT workflows in Python and SQL, streamlining data acquisition and improving operational efficiency by 30%.
- Integrated SAP source systems into enterprise data platforms using ETL/ELT pipelines.
- Optimized complex SQL queries and transformations, reducing runtime by 25% across analytics workloads.
- Partnered with analytics and clinical teams to design data models and cleansing processes, improving data accuracy and trust by 25%.
- Collaborated with cross-functional teams to deliver enterprise data warehouse solutions, increasing data accessibility by 35%.
- Led and mentored a team of 5 data engineers, improving data quality metrics by 30% and promoting best practices in cloud engineering and governance.
- Supported predictive analytics initiatives, including patient readmission models, enabling improved care outcomes and risk stratification.
- Designed and maintained Databricks lakehouse architectures for large-scale healthcare analytics and ML workloads.
- Developed and maintained dbt models to standardize transformations, enforce data quality, and enable analytics-ready datasets.
- Built Power BI dashboards to support executive, clinical, and operational reporting with real-time and batch data.
- Partnered with stakeholders to translate business and clinical requirements into actionable visualizations.
- Designed data workflows using Microsoft Fabric, integrating lakehouse storage, data engineering, and analytics capabilities.
- Designed and maintained Airflow DAGs to orchestrate complex data pipelines and dependencies.
- Developed Tableau dashboards for healthcare operations, finance, and clinical performance tracking.

Senior Data Engineer

Feb 2017 - Aug 2021

Curve Dental

Responsibilities:

- Architected a secure, serverless data pipeline using AWS Glue and Lambda, processing 2.2M+ healthcare records per month with 99.9% uptime.
- Migrated legacy on-prem ETL workflows to Snowflake, achieving a 38% reduction in query response times.
- Built and operated FHIR-based data conversion services, processing 1.5M records daily to support statewide health information exchange.
- Integrated Epic EHR data into enterprise data platforms for analytics, reporting, and clinical insights.
- Developed Looker/LookML models to centralize business logic and ensure consistent metric definitions.
- Developed robust ETL pipelines in Python and SQL to ingest EHR and claims data from 30+ hospital systems.
- Implemented Change Data Capture (CDC) pipelines, reducing data latency from 24 hours to under 2 hours.
- Introduced automated data quality validation using Great Expectations, reducing error rates from 2.4% to 0.3%.
- Designed and maintained a centralized data catalog, improving dataset discoverability for 200+ analysts and stakeholders.
- Implemented dbt tests, documentation, and lineage, improving trust and transparency across analytics teams.
- Automated BI and reporting workflows using Tableau, reducing manual reporting effort by 80%.
- Collaborated closely with product and analytics teams on predictive modeling initiatives, improving clinical risk stratification accuracy by 12%.
- Optimized Spark jobs on Databricks using partitioning, caching, and cluster tuning to reduce runtime and cost.
- Participated in HIPAA audits and compliance reviews, ensuring all pipelines met regulatory and security standards.
- Mentored 3 junior engineers, strengthening team expertise in cloud data engineering and healthcare compliance.

Data Engineer

Sep 2014 - Feb 2017

Miratech

Responsibilities:

- Designed and implemented data-driven software solutions aligned with customer and business requirements.
- Contributed across the full SDLC, including system design, documentation, development, testing, deployment, and support.
- Developed and enhanced ETL systems using Python and .NET, improving data reliability and performance.
- Advised on technology selection and architecture decisions to ensure scalable and maintainable solutions.
- Proactively evaluated emerging technologies to improve system efficiency and long-term maintainability.
- Developed Qlik dashboards and data models for operational and executive analytics.
- Integrated Qlik with cloud data warehouses to support near-real-time reporting
- Used Kibana to visualize logs, metrics, and pipeline health for observability and troubleshooting.